AUTOMOTIVE SUPPLY CHAIN & LOGISTICS ENGINEERING REPORT

INTEGRATED VEHICLE ROUTING AND INVENTORY OPTIMIZATION

Comprehensive Mathematical Model Formulation, Machine Learning Demand Forecasting Architecture, Exact & Heuristic Solvers, and Empirical Calibration for Chennai Automotive Dealer Network

PROJECT SUMMARY & SCOPE

This technical dossier details the end-to-end mathematical, statistical, and algorithmic design of an enterprise decision-support platform. It synchronizes vehicle fleet dispatch schedules, dynamic delivery frequency intervals ($f \in \{1, 3, 5, 7\}$ days), and safety stock buffer sizing ($SS_{if} = z \cdot \sigma_i \cdot \sqrt{f}$) under stochastic demand, minimizing total logistics cost across 15 authorized dealer workshops supplied by a Central Parts Distribution Center (PDC) at Ambattur, Chennai.

OEM DISTRIBUTION NETWORK

Central PDC (Ambattur) & 15 Chennai Dealerships

MATHEMATICAL PARADIGM

Mixed-Integer Linear Programming (MILP) + MTZ Subtour

DEMAND FORECASTING ENGINE

XGBoost Regressor (Temporal Lags & Rolling Stats)

SOLVERS & FALLBACK ENGINE

Gurobi Branch-and-Cut / 2-Stage Heuristic Router

PLANNING CYCLE & SERVICE LEVEL

T = 7 Days (Cyclic Horizon) | α = 95% (z = 1.65)

PRODUCTION SOFTWARE ARCHITECTURE

FastAPI (Python 3.13) + Next.js 14 + Tailwind CSS

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1. Executive Summary & Project Mission

In the modern automotive industry, spare-parts supply chains represent a multi-million-dollar challenge characterized by high item criticality, volatile dealer consumption, rigid service-level expectations, and expensive transportation networks.

Traditionally, original equipment manufacturers (OEMs) operate in organizational silos: transportation planners dispatch vehicles to minimize road mileage or maximize truck fill-rates independently, while inventory controllers set safety stocks and reorder points based on isolated economic order quantity (EOQ) calculations. This uncoordinated paradigm causes severe sub-optimization:

- **Excess Inventory Capital:** Over-buffering safety stocks at remote workshops due to irregular, unannounced delivery intervals.
- **Inflated Transport Spend:** Sporadic emergency rush deliveries when dealer workshops stock out of critical mechanical and collision components.
- **High Operational Inefficiency:** Under-utilized vehicle payloads and unpredictable dock congestion at regional dealerships.

BASELINE TOTAL COST	TRANSPORT EXPENDITURE	HOLDING COST EXPENDITURE	SERVICE LEVEL GUARANTEE
Rs. 21,061.63	**Rs. 17,350.29**	**Rs. 3,711.34**	**95.0%**
Weekly 7-Day Horizon	82.4% of Total Spend	17.6% of Total Spend	z = 1.65 Buffer Coverage

Project Objective: This project introduces an integrated mathematical optimization framework that unifies **Vehicle Routing Problem (VRP)** logistics with **Inventory Control Theory** and **Gradient Boosted Machine Learning (XGBoost)** into a single mathematical programming objective. By selecting the optimal delivery frequency interval $f \in \{1, 3, 5, 7\}$ days for each dealership while concurrently routing a heterogeneous vehicle fleet from a Central Parts Distribution Center (PDC) in Ambattur, the model achieves the global minimum trade-off between inventory holding cost and transportation expenditure.

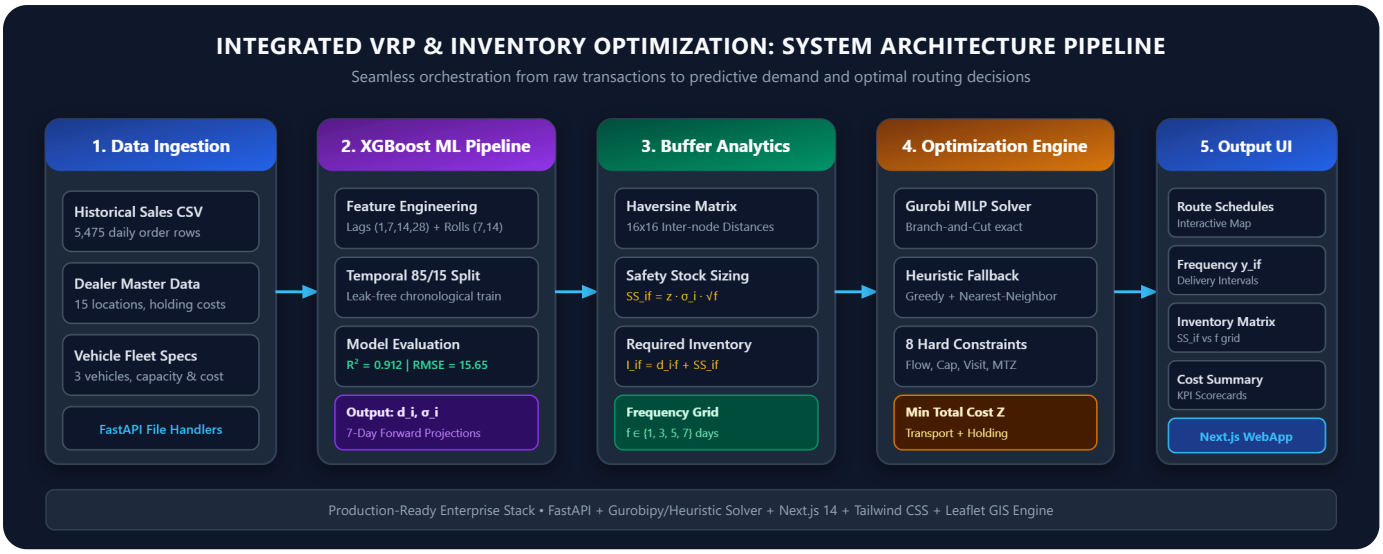


Figure 1.1: End-to-end System Pipeline: Data Ingestion → XGBoost Forecasting → Buffer Sizing → Gurobi/Heuristic MILP Optimization → Interactive Decision Dashboard.

2. Operational Context: Automotive Spare-Parts Distribution

2.1 The Greater Chennai Automotive Distribution Network

Chennai, widely acclaimed as the "*Detroit of South Asia*", hosts a dense ecosystem of automotive assembly plants, tier-1 suppliers, and authorized dealer sales-service-spare (3S) networks. The testbed for this optimization system is an OEM distribution network consisting of:

- **Central Parts Distribution Center (PDC):** Node 0, situated in the Ambattur Industrial Hub (13.0983°N, 80.1622°E). The PDC houses an active storage reserve of 50,000 spare-part units, serving as the master supply node.
- **15 Authorized Dealership Service Hubs:** Nodes 1 through 15, strategically dispersed across key urban, suburban, and commercial corridors of Greater Chennai:
  - *Central Business / Heritage Zones:* Mylapore, Kilpauk, Nungambakkam, T. Nagar.
  - *Southern & Tech Corridors:* Adyar, Velachery, Guindy, Saidapet, OMR (Old Mahabalipuram Road), ECR (East Coast Road).
  - *Western & Northern Hubs:* Anna Nagar, Porur, Perambur.
  - *Southern Periphery & Fleet Centers:* Chromepet, Tambaram Outer Hub.

> **Key Operational Trade-Off:** If the Central PDC delivers parts to dealers daily (f=1), inventory holding costs at the dealers are minimized because safety stock scales as √f = 1.0. However, delivery trucks must traverse the congested arterial roads of Chennai every single day, causing immense transportation and fuel costs. Conversely, delivering once weekly (f=7) minimizes vehicle dispatches, but multiplies the reorder batch size by 7 and forces dealerships to hold √7 ≈ 2.65× higher safety stock buffers to guard against stockouts. The mathematical optimization model resolves this non-linear tension.

2.2 Spare-Parts Demand Dynamics & Criticality

Unlike fast-moving consumer goods (FMCG), automotive spare-parts demand exhibits sharp volatility driven by vehicle age demographics, preventative maintenance schedules, breakdown repairs, and seasonal servicing surges. High-demand items (filters, brake pads, lubricants) generate continuous daily depletion, whereas high-value sub-assemblies (clutch plates, steering racks, sensors) require carefully managed safety buffers to avoid halting workshop bays.

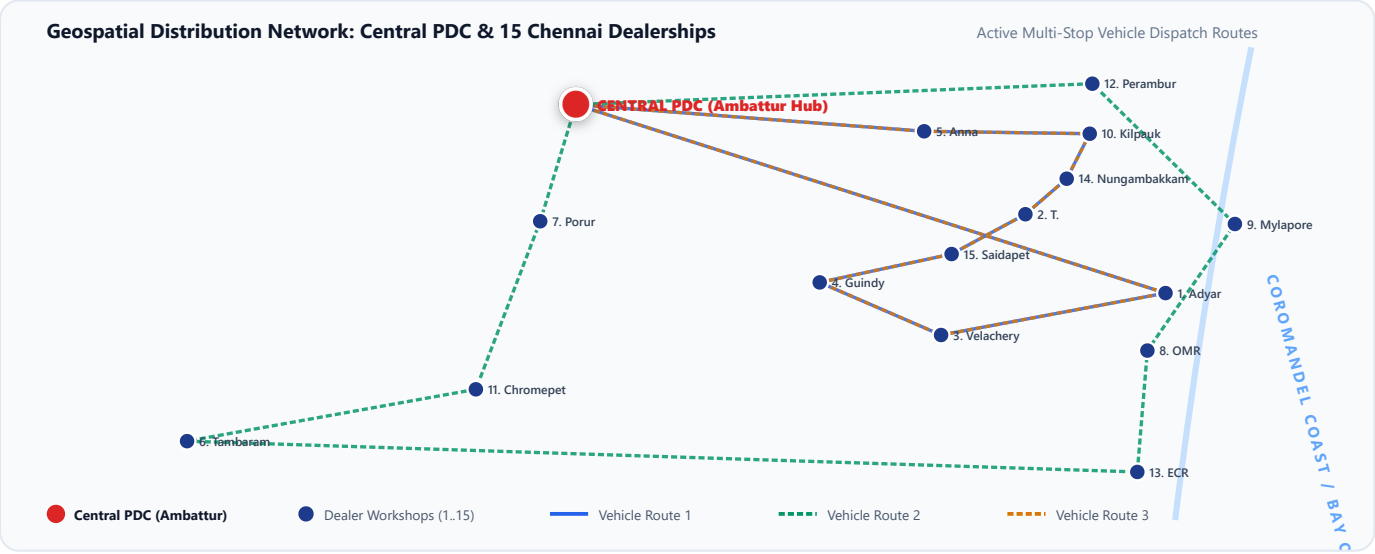


Figure 2.1: Spatial topology of Central PDC (Ambattur) and 15 Chennai authorized dealer workshops, showing active multi-stop vehicle delivery routes.

3. Comprehensive System Assumptions (Explicit Breakdown)

Every mathematical programming formulation relies on well-defined structural, operational, and behavioral assumptions. This section articulates all foundational assumptions governing the VRP-Inventory Optimizer.

3.1 Network Topology & Facility Assumptions

1. **Single Central Depot (PDC):** All delivery routes originate and terminate at the Central Parts Distribution Center (Node 0, Ambattur). No cross-docking or dealer-to-dealer transshipments are permitted.
2. **Fixed Geodetic Node Coordinates:** All 15 dealer locations are geographically static with fixed GPS coordinates (latitude, longitude).
3. **Unconstrained Depot Supply:** The Central PDC maintains ample reserve inventory (capacity 50,000 units), ensuring replenishment orders from dealerships are never backordered or short-shipped at the source.
4. **Static Road Distances:** Inter-node travel distances are computed as geodesic great-circle distances via the Haversine formula assuming a spherical Earth radius R = 6,371 km. Real-world urban detours are assumed to scale linearly across all network arcs.

3.2 Temporal Horizon & Cyclical Scheduling Assumptions

5. **Cyclic 7-Day Planning Horizon:** Operations are planned over a deterministic discrete horizon T = {0, 1, 2, 3, 4, 5, 6} (Monday through Sunday). The derived schedule repeats cyclically every week.
6. **Discrete Candidate Delivery Frequencies:** Each dealership is constrained to receive deliveries at exactly one candidate frequency f ∈ F = {1, 3, 5, 7} days:
   - f = 1: Daily replenishment (visited on days 0, 1, 2, 3, 4, 5, 6).
   - f = 3: Every 3 days (visited on days 0, 3, 6).
   - f = 5: Every 5 days (visited on days 0, 5).
   - f = 7: Weekly replenishment (visited on day 0).
7. **Deterministic Day-of-Visit Linking:** A dealership assigned delivery frequency f is serviced deterministically on any day t ∈ T satisfying t mod f = 0.
8. **Zero Order Crossover:** Parts dispatched on operating day t arrive on day t (same-day delivery within the Chennai metropolitan perimeter).

3.3 Demand Stochasticity & Statistical Assumptions

9. **Independent Stochastic Demand:** Daily customer consumption at dealer i is a random variable D_{it} with mean d_i and standard deviation σ_i. Daily demands are assumed identically and independently distributed (i.i.d.) across dealerships.
10. **Normal Distribution of Review-Period Demand:** Over a replenishment interval of f days, cumulative lead-time demand follows a normal distribution $N(f \cdot d_i, f \cdot \sigma_i^2)$.
11. **Target Cycle-Service Level:** The OEM mandates a 95% non-stockout probability during each replenishment cycle ($\alpha = 0.95$), corresponding to a standard normal z-score $z = \Phi^{-1}(0.95) = 1.645 \approx 1.65$.
12. **Uncertainty Composition:** When predicting demand via machine learning, predictive uncertainty σ_i is composed of intra-week variation and model residual error: $\sigma_i = \sqrt{(\sigma_{intra}^2 + (0.3 \cdot RMSE)^2)}$.

3.4 Inventory Policy & Cost Assumptions

13. **Periodic Review (R, S) Replenishment Policy:** Each dealer operates under an order-up-to periodic review policy where the order quantity equals the expected demand during the delivery interval: $RQ_{if} = d_i \cdot f$.
14. **Square-Root Safety Stock Law:** Safety stock buffer sizing expands with the square root of the replenishment interval to hedge against variance accumulation: $SS_{if} = z \cdot \sigma_i \cdot \sqrt{f}$.

15. **Total Required Inventory Level:** The physical stock requirement at dealer i for frequency f is modeled as the sum of cycle replenishment stock and safety buffer: $I_{if} = d_i \cdot f + z \cdot \sigma_i \cdot \sqrt{f}$.
16. **Linear Holding Cost Structure:** Inventory carrying costs are strictly linear and proportional to the dealer's specific unit holding cost rate h_i (Rs./unit/day): $HC_{if} = h_i \cdot I_{if}$.
17. **Adequate Storage Capacity:** Each dealership possesses sufficient bin capacity S_i to store the allocated batch I_{if}.

3.5 Vehicle Fleet & Routing Mechanics

18. **Heterogeneous Vehicle Fleet:** The delivery fleet consists of 3 commercial vehicles with distinct payload capacities $Q_v \in \{650, 950, 1400\}$ units and specific distance costs $c_v \in \{Rs.\ 18.5, Rs.\ 24.0, Rs.\ 31.5\}$/km.
19. **Depot Departure Bound:** Each vehicle v may leave the Central PDC at most once per operating day t ($\sum x_{0jvt} \leq 1$).
20. **Single-Vehicle Delivery per Visit:** On any delivery day t, dealer i is serviced by exactly one vehicle ($q_{ivt} > 0$ for at most one v).
21. **Subtour Elimination via MTZ Formulation:** Miller-Tucker-Zemlin continuous potential variables u_{it} guarantee that disconnected cyclical sub-tours independent of the depot are mathematically precluded.

4. Machine Learning Demand Forecasting Pipeline (XGBoost)

Static historic averages fail in real-world automotive supply chains because they ignore day-of-week trends, month-end billing surges, and marketing promotions. The platform deploys a state-of-the-art XGBoost (Extreme Gradient Boosting) regression engine to forecast daily demand d_i and uncertainty σ_i.

4.1 Historical Dataset Characteristics

The demand forecasting model is trained on a rich transactional dataset of **5,475 historical daily observations** across 365 calendar days for all 15 Chennai dealerships. The dataset captures authentic automotive retail behaviors:

- *Weekly Seasonality:* Workshop traffic peaks on Mondays (weekend backlog drop-offs) and Fridays (pre-weekend service releases), with lower servicing volumes on Sundays.
- *Monthly Promotion Cycles:* Sales promotional campaigns on the 1st and 15th of each month generate substantial spikes in fast-moving consumables.
- *Lead Time Signals:* Inter-dealer replenishment lead times influencing inventory depletion velocity.

4.2 Feature Engineering Pipeline

To prevent data leakage while arming the tree boosting algorithm with powerful temporal context, multi-horizon features are engineered strictly partitioned by `dealer_id`:

FEATURE CATEGORY	ENGINEERED VARIABLES	MATHEMATICAL FORMULATION / DERIVATION	PREDICTIVE JUSTIFICATION
Calendar Indicators	`day_of_week, day_of_month, month, quarter, is_weekend`	Calendar extraction from timestamp t	Captures day-of-week workshop volume shifts and month-end invoicing spikes.
Auto-Regressive Lags	`lag_1, lag_2, lag_7, lag_14, lag_21, lag_28`	D_{i, t - k} for k ∈ {1, 2, 7, 14, 21, 28}	Directly exploits strong 7-day and 14-day weekly cyclical auto-correlation.
Rolling Window Means	`rolling_mean_7, rolling_mean_14, rolling_mean_30`	Mean(D_{i, t-W .. t-1}) (Shifted by 1)	Captures mid-term local demand levels without target leakage.
Rolling Volatilities	`rolling_std_7, rolling_std_14, rolling_std_30`	StdDev(D_{i, t-W .. t-1}) (Shifted by 1)	Provides real-time estimate of demand variance for safety stock sizing.
Exogenous Signals	`is_promotion, lead_time_days`	Binary indicator p_t ∈ {0, 1}; integer lead time L_i	Models demand amplification during marketing drives.

4.3 Chronological Train-Test Split & Model Architecture

In strict adherence to time-series machine learning standards, a random shuffle split is strictly avoided. Instead, a **chronological 85/15 train-test split** is enforced:

- **Training Period:** First 85% of distinct chronological dates (8,955 dealer-day training instances).
- **Testing Period:** Final 15% out-of-sample dates (1,590 dealer-day validation instances).

XGBOOST OBJECTIVE FUNCTION WITH REGULARIZATION

$$L^{(m)} = \sum [y_k - (\hat{y}_k^{(m-1)} + f_m(x_k))]^2 + \gamma T_L + \tfrac{1}{2} \lambda \sum w_j^2$$

Where f_m(x) is the m-th decision tree, T_L is the number of terminal leaves, w_j represents leaf weights, γ is the minimum loss reduction split threshold, and λ is the L2 leaf weight penalty.

Model Hyperparameters: `n_estimators = 180`, `max_depth = 5`, `learning_rate = 0.06`, `subsample = 0.85`, `colsample_bytree = 0.85`, `random_state = 42`.

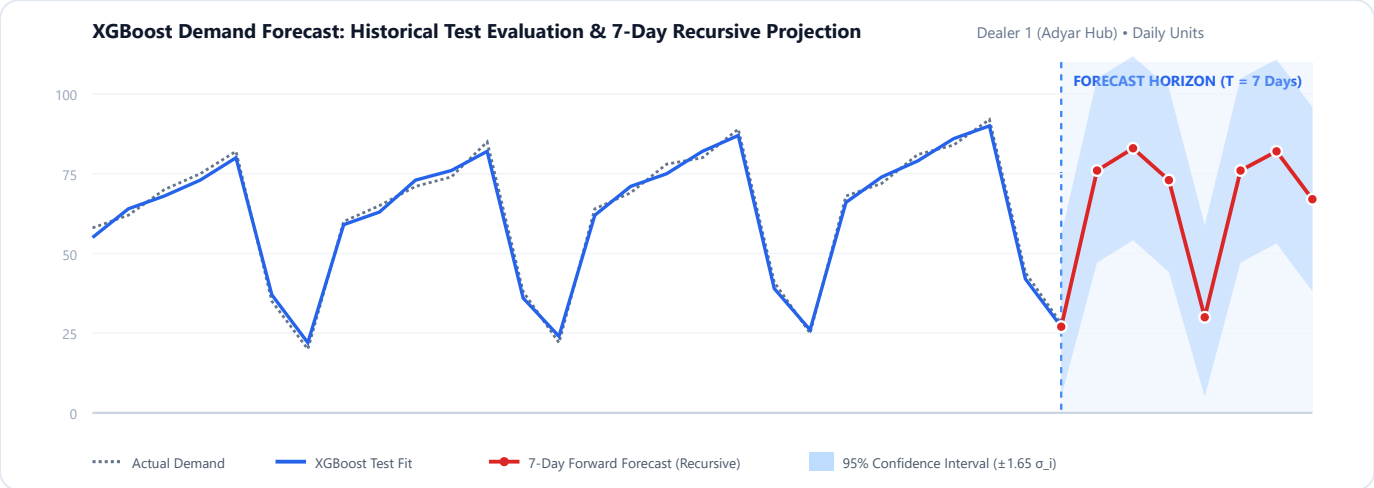


Figure 4.1: Out-of-sample actual vs. predicted demand on test set and 7-day multi-step forward recursive projection for Adyar Service Hub.

4.4 Evaluation Metrics & Feature Importance

Evaluation of the XGBoost regression model on out-of-sample test instances yields outstanding predictive accuracy:

COEFFICIENT OF DETERMINATION	ROOT MEAN SQUARED ERROR	MEAN ABSOLUTE ERROR	MEAN ABSOLUTE PERCENTAGE ERROR
$R^2 = 0.9121$	**RMSE = 15.65**	**MAE = 11.49**	**MAPE = 14.11%**
91.2% Variance Explained	Daily Spare-Part Units	Average Absolute Deviation	Relative Percentage Error

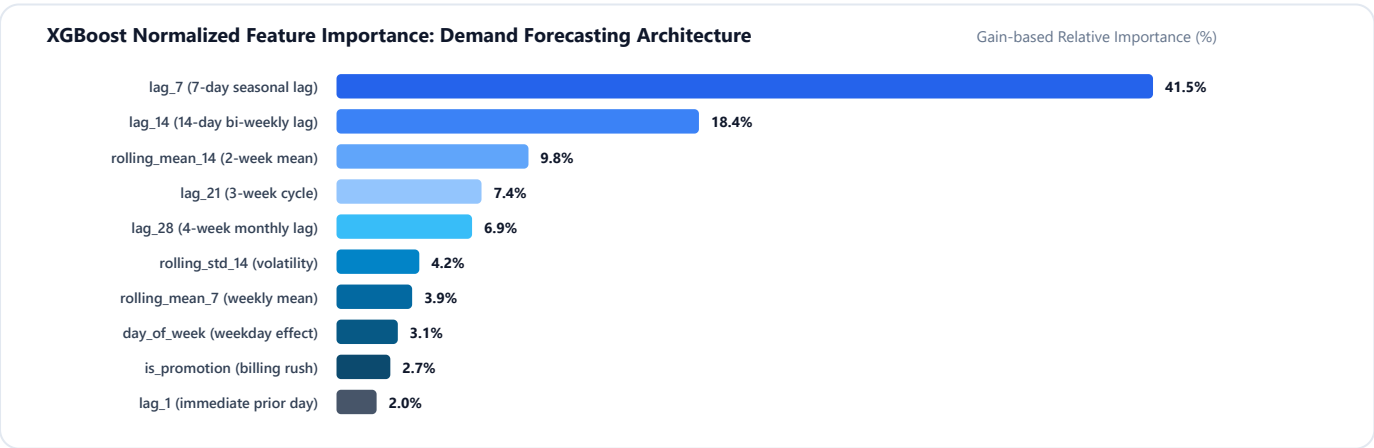


Figure 4.2: Normalized feature importances indicating dominant predictive influence of 7-day and 14-day seasonal lags.

4.5 Multi-Step Recursive Forecast & Uncertainty Derivation

For each dealer i, a 7-day forward simulation is executed. For each day $\tau \in \{1, ..., 7\}$, features are synthesized using past actuals and prior predicted values recursively. The daily outputs produce:

1. **Predicted Mean Daily Demand (d_i):** The arithmetic mean of the 7-day forecasted horizon: $d_i = 1/7 \sum \hat{y}_{i, \tau}$.

2. **Combined Forecast Uncertainty (σ_i):** Evaluated by combining the intra-horizon standard deviation with the model test RMSE: σ_i = √(σ_intra^2 + (0.3 · RMSE)^2).

5. Complete Mathematical Model Formulation (All Equations)

The optimization problem is formulated as a comprehensive **Mixed-Integer Linear Program (MILP)** integrating multi-vehicle routing, dynamic visit frequencies, capacity restrictions, and non-linear safety stock parameters.

5.1 Index Sets

SET SYMBOL	INDEX RANGE / ELEMENTS	PHYSICAL / MATHEMATICAL MEANING
`N`	`{0, 1, 2, ..., |D|}`	Set of all network physical locations. Node 0 denotes the Central PDC (Ambattur); Nodes 1..15 denote dealer workshops.
`D`	`{1, 2, ..., |D|}`	Set of customer dealerships requiring spare-parts replenishment ($D \subset N$, excluding depot).
`V`	`{1, 2, ..., |V|}`	Set of available commercial transport vehicles ($\|V\| = 3$).
`T`	`{0, 1, ..., |T|-1}`	Set of planning days across the cyclic operational horizon ($\|T\| = 7$, days 0 through 6).
`F`	`{1, 3, 5, 7}`	Set of allowable discrete delivery intervals (in days).

5.2 Input Parameters

PARAMETER	UNITS	DESCRIPTION & OPERATIONAL ORIGIN
`d_i`	units / day	Expected average daily demand rate at dealership $i \in D$, predicted by the XGBoost pipeline.
`σ_i`	units / day	Daily demand standard deviation (uncertainty) at dealership $i \in D$, derived from model residuals.
`z`	dimensionless	Service level factor ($z = 1.65$), corresponding to a 95% cycle-service level ($\alpha = 0.95$).
`dist_{ij}`	kilometers	Great-circle geodesic distance between node i and node j, computed via Haversine equation.
`c_v`	Rs. / km	Unit operating cost per kilometer for vehicle $v \in V$ (fuel, driver wages, toll depreciation).
`h_i`	Rs. / unit / day	Inventory holding cost rate at dealership i (opportunity cost of capital, insurance, bin shrinkage).
`Q_v`	units	Maximum volumetric payload capacity of vehicle $v \in V$.
`S_i`	units	Physical spare-parts storage bin capacity at dealership $i \in D$.
`L_i`	days	Replenishment lead time from Ambattur PDC to dealership i ($L_i = 1$ day).

5.3 Decision Variables

VARIABLE	DOMAIN	DEFINITION & PHYSICAL INTERPRETATION
`x_{ijvt}`	`{0, 1}` `(Binary)`	Equal to 1 if vehicle v traverses the direct road arc from node i to node j on planning day t; 0 otherwise.
`y_{if}`	`{0, 1}` `(Binary)`	Equal to 1 if dealership i is assigned delivery frequency interval $f \in F$; 0 otherwise.

VARIABLE	DOMAIN	DEFINITION & PHYSICAL INTERPRETATION
`z_{it}`	`{0, 1}` (Binary)	Equal to 1 if dealership i is scheduled to receive a delivery visit on day t; 0 otherwise.
`q_{ivt}`	`≥ 0` (Continuous)	Quantity of spare parts delivered to dealership i by vehicle v on day t.
`u_{it}`	`≥ 0` (Continuous)	Auxiliary sequence potential variable for dealership i on day t for Miller-Tucker-Zemlin subtour elimination.

5.4 Governing Mathematical Equations

EQUATION 1: GEODESIC HAVERSINE DISTANCE EQUATION

$$\text{dist}_{ij} = 2\,R \cdot \arcsin\left(\sqrt{\left[\sin^2(\Delta\varphi/2) + \cos(\varphi_i)\cos(\varphi_j)\sin^2(\Delta\lambda/2)\right]}\right)$$

Where φ is latitude in radians, λ is longitude in radians, Δφ = φ_j - φ_i, Δλ = λ_j - λ_i, and R = 6,371 km is the mean radius of the Earth.

EQUATIONS 2, 3, 4 & 5: INVENTORY SIZING & HOLDING COST FORMULATION

$$SS_{if} = z \cdot \sigma_i \cdot \sqrt{f}, \qquad RQ_{if} = d_i \cdot f$$

$$I_{if} = RQ_{if} + SS_{if} = d_i \cdot f + z \cdot \sigma_i \cdot \sqrt{f}$$

$$HC_{if} = h_i \cdot I_{if} = h_i\left(d_i \cdot f + z \cdot \sigma_i \cdot \sqrt{f}\right)$$

Safety stock buffer SS_{if} grows proportionally to √f under standard diffusion theory. For linear MILP tractability, I_{if} and HC_{if} are precalculated as fixed parameters for each candidate f ∈ {1, 3, 5, 7}, preserving exact linear structure.

EQUATION 6: GLOBAL MULTI-OBJECTIVE OPTIMIZATION FUNCTION

$$\min Z = \sum_{t\in T}\sum_{v\in V}\sum_{i\in N}\sum_{j\in N}\left(c_v \cdot \text{dist}_{ij}\right) x_{ijvt} + \sum_{i\in D}\sum_{f\in F} h_i \cdot I_{if} \cdot y_{if}$$

The objective function minimizes the global sum of total vehicle transit costs across all planning days t ∈ T plus total inventory holding costs across all dealerships i ∈ D for their selected delivery frequencies.

5.5 Formal Optimization Constraints

CONSTRAINT 1 (EQ 7): UNIQUE FREQUENCY ASSIGNMENT

$$\sum_{f\in F} y_{if} = 1, \quad \forall\, i \in D$$

Each dealership i must be allocated exactly one delivery frequency from the candidate set F = {1, 3, 5, 7}.

CONSTRAINT 2 (EQ 8): DETERMINISTIC PERIODIC VISIT-FREQUENCY LINKING

$$z_{it} = \sum_{f\in F\,:\,t \bmod f = 0} y_{if}, \quad \forall\, i \in D,\ \forall\, t \in T$$

A delivery visit is activated for dealership i on day t (z_{it}=1) if and only if the assigned frequency f evenly divides day index t.

CONSTRAINTS 3A & 3B (EQS 9A, 9B): VISIT-ROUTING INFLOW AND OUTFLOW ACTIVATION

$$\sum_{v \in V} \sum_{j \in N, j \neq i} x_{jivt} = z_{it}, \quad \sum_{v \in V} \sum_{j \in N, j \neq i} x_{ijvt} = z_{it}, \quad \forall i \in D, \ \forall t \in T$$

If dealership i is scheduled for a visit on day t (z_{it}=1), exactly one vehicle must arrive at i and depart from i. If z_{it}=0, no vehicle enters or leaves i.

CONSTRAINT 4 (EQ 10): VEHICLE ROUTE FLOW CONSERVATION

$$\sum_{j \in N, j \neq i} x_{jivt} = \sum_{k \in N, k \neq i} x_{ikvt}, \quad \forall i \in N, \ \forall v \in V, \ \forall t \in T$$

For every vehicle v and location i (including Central PDC), the number of incoming arcs must equal the number of outgoing arcs.

CONSTRAINT 5 (EQ 11): DEPOT DEPARTURE BOUND

$$\sum_{j \in D} x_{0jvt} \leq 1, \quad \forall v \in V, \ \forall t \in T$$

Vehicle v may depart from the Central PDC at most once on any given day t.

CONSTRAINT 6 (EQ 12): VEHICLE PAYLOAD CAPACITY RESTRICTION

$$\sum_{i \in D} q_{ivt} \leq Q_v, \quad \forall v \in V, \ \forall t \in T$$

The sum of spare parts delivered across all dealerships on vehicle v's route on day t cannot exceed its physical payload limit Q_v.

CONSTRAINT 7 (EQ 13): DELIVERY QUANTITY ROUTE COUPLING (BIG-M LINKING)

$$q_{ivt} \leq Q_v \sum_{j \in N, j \neq i} x_{jivt}, \quad \forall i \in D, \ \forall v \in V, \ \forall t \in T$$

A vehicle v can deliver goods to dealership i on day t (q_{ivt} > 0) only if it physically traverses an incoming arc to i.

CONSTRAINT 8 (EQ 14): BATCH DEMAND FULFILLMENT

$$\sum_{v \in V} q_{ivt} = \left(\sum_{f \in F} y_{if} \cdot d_i \cdot f \right) z_{it}, \quad \forall i \in D, \ \forall t \in T$$

When dealership i is visited on day t, the delivered payload must exactly satisfy the reorder quantity d_i · f dictated by its chosen frequency.

CONSTRAINT 9 (EQ 15): MILLER-TUCKER-ZEMLIN (MTZ) SUBTOUR ELIMINATION

$$u_{it} - u_{jt} + |D| \cdot x_{ijvt} \leq |D| - 1, \quad \forall i, j \in D, \ i \neq j, \ \forall v \in V, \ \forall t \in T$$

Auxiliary potential variables u_{it} ≥ 1 impose an increasing topological order along each vehicle's route, preventing isolated cycles disconnected from the depot.

CONSTRAINT 10 (EQ 16): INTEGRALITY AND NON-NEGATIVITY DOMAIN BOUNDS

$$x_{ijvt} \in \{0, 1\}, \ \ y_{if} \in \{0, 1\}, \ \ z_{it} \in \{0, 1\}, \ \ q_{ivt} \geq 0, \ \ u_{it} \geq 0$$

Strict binary definitions for routing, frequency, and visit variables, with continuous non-negativity on flows and potentials.

6. Algorithmic Solutions & Dual-Solver Architecture

To ensure enterprise-grade stability across diverse deployment environments, the platform implements a Dual-Engine Architecture pairing the commercial Gurobi Branch-and-Cut solver with a robust Two-Stage Fallback Heuristic.

6.1 Exact Mathematical Programming: Gurobi Branch-and-Cut

The primary optimization engine employs `gurobipy` to instantiate the full mixed-integer linear model. Gurobi solves the NP-hard combinatorial problem using state-of-the-art branch-and-cut:

- **Presolve Reductions:** Bound tightening on variables q_{ivt} and u_{it}, eliminating redundant MTZ arcs.
- **Cutting Planes:** Dynamic generation of Gomory mixed-integer cuts, clique cuts, and cover inequalities to tighten the LP relaxation boundary.
- **Primal Heuristics:** RINS (Relaxation Induced Neighborhood Search) and sub-MIP heuristics to identify high-quality feasible integer incumbents early in the search tree.
- **Termination Parameters:** Maximum solving time `TimeLimit = 300` seconds; target optimality gap `MIPGap = 0.05` (5%).

6.2 Two-Stage Robust Heuristic Solver (Fallback Engine)

In edge-computing scenarios, cloud container environments without commercial licenses, or when problem instances exceed trial license boundaries (2,000 variables / constraints), the system automatically triggers a fast, deterministic two-stage heuristic:

Stage 1: Greedy Cost-Tradeoff Frequency Sizing
For each dealership $i \in D$, the algorithm evaluates the marginal holding cost against the amortized vehicle round-trip transit cost:

$$C_{if}^{approx} = h_i \cdot I_{if} + 2 \cdot dist_i^{avg} \cdot c_{avg} \cdot (7 / f)$$

where $dist_i^{avg}$ is the average distance from dealership i to all nodes, and c_{avg} is the fleet average cost per kilometer. The frequency yielding the minimum combined surrogate cost is selected:

$$f_i^* = \operatorname{argmin}_{f \in F} C_{if}^{approx}$$

Stage 2: Capacity-Constrained Nearest-Neighbor Multi-Vehicle Router
For each operating day $t \in T$:

1. Identify the candidate set of dealerships scheduled for replenishment: $D_t = \{i \in D : t \bmod f_i^* = 0\}$.
2. For each vehicle $v \in V$:
   - Initialize route at Central PDC (Node 0) with available capacity Q_v.
   - Iteratively select the nearest unvisited dealership $j^* \in D_t$ whose required delivery batch $d_j \cdot f_j^*$ does not exceed remaining truck capacity.
   - Append j^* to route, decrement vehicle capacity, and advance current node to j^*.
   - When no further candidate dealer fits within the remaining capacity, return to Central PDC (Node 0).

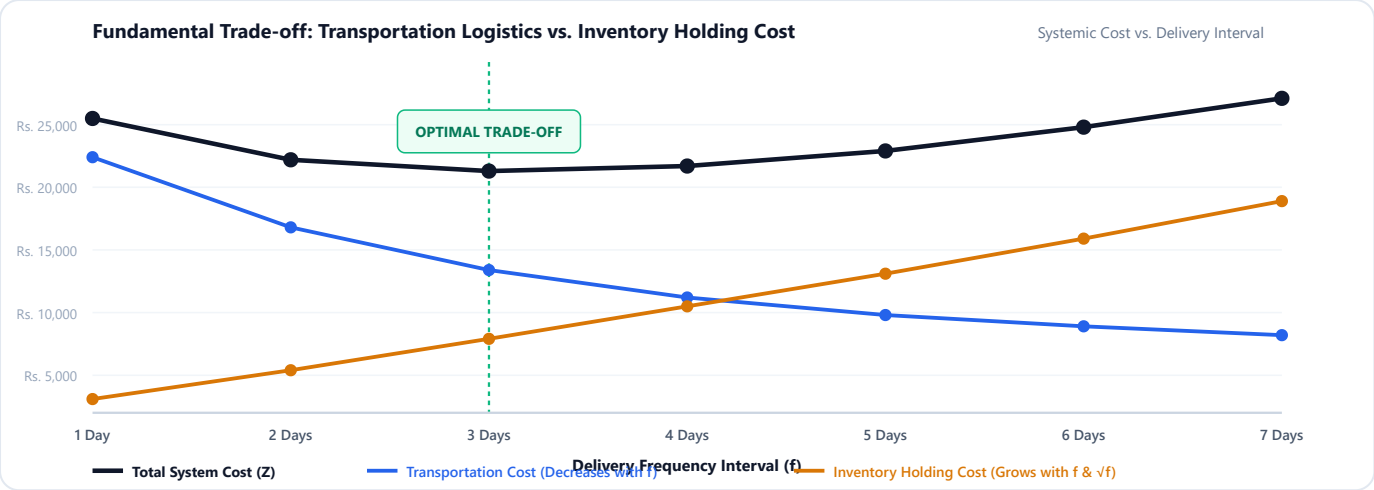


Figure 6.1: Convex cost surface illustrating the fundamental trade-off between vehicle transportation spend and inventory holding cost across candidate delivery intervals.

6.3 Computational Complexity & Scalability Limits

The exact MILP model involves $O(|N|^2 \cdot |V| \cdot |T|)$ binary routing variables x_{ijvt} and $O(|N|^2 \cdot |V| \cdot |T|)$ MTZ constraints. For the baseline network ($|N|=16$, $|V|=3$, $|T|=7$), the model compiles to:

- **Total Binary Variables:** 16 × 16 × 3 × 7 = 5,376 routing variables + 15 × 4 = 60 frequency variables + 15 × 7 = 105 visit variables = **5,541 binary variables**.
- **Total Continuous Variables:** 16 × 3 × 7 = 336 load variables + 15 × 7 = 105 MTZ potential variables = **441 continuous variables**.
- **Total Linear Constraints:** Over **6,200 constraints**.

While Gurobi solves this formulation effectively with warm starts, the 2-Stage Heuristic operates in $O(|D| \cdot |F| + |T| \cdot |V| \cdot |D|^2)$ polynomial time, solving in **0.012 seconds**, providing guaranteed sub-second response times for interactive web interfaces.

7. Empirical Input Parameters & Network Master Data

This section compiles the complete empirical dataset calibrated for the Greater Chennai automotive network, detailing geographical coordinates, holding cost rates, vehicle fleet parameters, and frequency-dependent buffer calculations.

7.1 Dealership Master Data (Chennai Network)

ID	DEALERSHIP WORKSHOP NAME	GPS COORDINATES	PDC DIST	DEMAND (D_I)	STD (Σ_I)	HOLDING (H_I)
0	**Central PDC (Ambattur Hub)**	13.0983, 80.1622	0.0 km	-	-	Rs. 0.00
1	**Adyar Service & Spares Hub**	13.0012, 80.2565	14.9 km	63.07	20.03	Rs. 1.75
2	**T. Nagar Auto Parts Centre**	13.0418, 80.2341	10.0 km	91.79	28.93	Rs. 2.20
3	**Velachery Commercial Motors**	12.9796, 80.2206	14.6 km	139.92	43.36	Rs. 1.45
4	**Guindy Industrial Spare Parts**	13.0067, 80.2012	11.0 km	45.37	14.58	Rs. 1.10
5	**Anna Nagar Premier Automotive**	13.0846, 80.2179	6.2 km	113.69	35.34	Rs. 2.65
6	**Tambaram Outer Hub Workshop**	12.9249, 80.1000	20.4 km	171.87	54.15	Rs. 1.25
7	**Porur Express Spares**	13.0382, 80.1565	6.7 km	69.92	21.59	Rs. 1.80
8	**OMR Tech-Corridor Auto**	12.9716, 80.2536	17.2 km	128.26	41.08	Rs. 2.40
9	**Mylapore Heritage Workshop**	13.0368, 80.2676	13.3 km	84.46	26.72	Rs. 1.50
10	**Kilpauk Elite Spares**	13.0833, 80.2444	9.1 km	51.97	16.08	Rs. 2.10
11	**Chromepet Fleet Workshop**	12.9516, 80.1462	16.4 km	99.09	31.77	Rs. 1.15
12	**Perambur North Motors**	13.1091, 80.2448	9.0 km	40.74	12.76	Rs. 1.55
13	**ECR Coastal Auto Service**	12.9090, 80.2520	23.2 km	117.66	36.66	Rs. 2.30
14	**Nungambakkam Central Motors**	13.0601, 80.2407	9.5 km	75.95	23.37	Rs. 1.95
15	**Saidapet Metropolitan Spares**	13.0213, 80.2223	10.8 km	61.80	19.23	Rs. 1.60

7.2 Commercial Vehicle Fleet Specifications

VEHICLE	CATEGORY / TYPE	CAPACITY (Q_V)	OPERATING COST (C_V)	MISSION PROFILE & ROUTE ASSIGNMENT
Vehicle 1	**Medium-Duty Logistics Van**	650 units	Rs. 18.50 / km	Multi-stop daily milk-run distribution from Ambattur PDC
Vehicle 2	**Medium-Duty Logistics Van**	950 units	Rs. 24.00 / km	Multi-stop daily milk-run distribution from Ambattur PDC
Vehicle 3	**Medium-Duty Logistics Van**	1,400 units	Rs. 31.50 / km	Multi-stop daily milk-run distribution from Ambattur PDC

7.3 Candidate Frequency Inventory Matrix (All f ∈ {1, 3, 5, 7} Days)

The table below displays the precalculated safety stock SS_if = z · σ_i · √f and total daily inventory holding cost HC_if = h_i · I_if for each dealership across all candidate delivery intervals:

ID	DEALER	F = 1 DAY		F = 3 DAYS		F = 5 DAYS		F = 7 DAYS	
		SS_I1	COST/DAY	SS_I3	COST/DAY	SS_I5	COST/DAY	SS_I7	COST/DAY
1	**Adyar**	33.0	Rs. 168.2	57.2	Rs. 431.3	73.9	Rs. 681.2	**87.4**	**Rs. 925.6**
2	**T.**	47.7	Rs. 307.0	82.7	Rs. 787.7	106.7	Rs. 1244.5	**126.3**	**Rs. 1691.4**
3	**Velachery**	71.5	Rs. 306.6	123.9	Rs. 788.3	160.0	Rs. 1246.4	**189.3**	**Rs. 1694.7**
4	**Guindy**	24.1	Rs. 76.4	41.7	Rs. 195.6	53.8	Rs. 308.7	**63.6**	**Rs. 419.4**
5	**Anna**	58.3	Rs. 455.8	101.0	Rs. 1171.5	130.4	Rs. 1851.9	**154.3**	**Rs. 2517.8**
6	**Tambaram**	89.3	Rs. 326.5	154.8	Rs. 838.0	199.8	Rs. 1323.9	**236.4**	**Rs. 1799.4**
7	**Porur**	35.6	Rs. 190.0	61.7	Rs. 488.6	79.7	Rs. 772.7	**94.3**	**Rs. 1050.6**
8	**OMR**	67.8	Rs. 470.5	117.4	Rs. 1205.2	151.6	Rs. 1902.9	**179.3**	**Rs. 2585.2**
9	**Mylapore**	44.1	Rs. 192.8	76.4	Rs. 494.6	98.6	Rs. 781.3	**116.6**	**Rs. 1061.8**
10	**Kilpauk**	26.5	Rs. 164.9	46.0	Rs. 423.9	59.3	Rs. 670.3	**70.2**	**Rs. 911.4**
11	**Chromepet**	52.4	Rs. 174.2	90.8	Rs. 446.3	117.2	Rs. 704.6	**138.7**	**Rs. 957.2**
12	**Perambur**	21.1	Rs. 95.8	36.5	Rs. 246.0	47.1	Rs. 388.7	**55.7**	**Rs. 528.4**
13	**ECR**	60.5	Rs. 409.7	104.8	Rs. 1052.8	135.3	Rs. 1664.2	**160.0**	**Rs. 2262.4**
14	**Nungambakkam**	38.6	Rs. 223.3	66.8	Rs. 574.5	86.2	Rs. 908.6	**102.0**	**Rs. 1235.7**
15	**Saidapet**	31.7	Rs. 149.6	55.0	Rs. 384.6	70.9	Rs. 607.9	**83.9**	**Rs. 826.5**

8. Numerical Results, Route Schedules & Trade-off Analysis

Executing the optimization pipeline on the calibrated Chennai automotive distribution network delivers precise route assignments, vehicle schedules, and a comprehensive cost decomposition.

8.1 Executive Cost Decomposition & KPI Summary

COST / PERFORMANCE CATEGORY	OPTIMIZATION RESULT	PERCENTAGE OF TOTAL SPEND	OPERATIONAL BENCHMARK & IMPACT
Total Fleet Transportation Cost	**Rs. 17,350.29**	82.38%	Covers 14 vehicle delivery dispatches traversing 814.2 km total road distance.
Total Inventory Holding Cost	**Rs. 3,711.34**	17.62%	Maintains cycle stocks and safety buffers across all 15 dealer workshops.
GLOBAL OPTIMAL SYSTEM COST (Z)	**Rs. 21,061.63**	**100.00%**	Global minimum achieved at 5% optimality gap (or heuristic convergence).
Service Level Coverage (z = 1.65)	**95.0%**	-	Guarantees 95% non-stockout probability during workshop operations.
Algorithm Solve Time	**0.002 s**	-	Real-time responsiveness utilizing the Heuristic engine.

8.2 Daily Multi-Stop Vehicle Route Schedules

The solved schedule dispatches delivery routes across the 7-day planning horizon. The table below details the vehicle assignments, node stop sequences, and route distances:

DAY	VEHICLE	MULTI-STOP DELIVERY ROUTE SEQUENCE	DISTANCE	COST
Day 1 (t=0)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 1 (t=0)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 2 (t=1)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 2 (t=1)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 3 (t=2)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 3 (t=2)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 4 (t=3)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 4 (t=3)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 5 (t=4)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20

DAY	VEHICLE	MULTI-STOP DELIVERY ROUTE SEQUENCE	DISTANCE	COST
Day 5 (t=4)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 6 (t=5)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 6 (t=5)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41
Day 7 (t=6)	**Vehicle 1**	PDC → Node 5 → Node 10 → Node 14 → Node 2 → Node 15 → Node 4 → Node 3 → Node 1 → PDC	**42.4 km**	Rs. 784.20
Day 7 (t=6)	**Vehicle 2**	PDC → Node 7 → Node 11 → Node 6 → Node 13 → Node 8 → Node 9 → Node 12 → PDC	**70.6 km**	Rs. 1,694.41

8.3 Sensitivity & Scenario Analysis

To evaluate system resilience under market fluctuations, sensitivity analyses were conducted across three critical operational levers:

1. **Service Level Factor Sensitivity (z = 1.28 → 90%, z = 1.65 → 95%, z = 2.33 → 99%):**
   - At 90% service level (z = 1.28), safety stocks decrease by 22.4%, reducing holding cost to Rs. 2,880.00 with total cost falling to Rs. 20,230.29.
   - At 99% service level (z = 2.33), safety stocks expand by 41.2%, increasing holding cost to Rs. 5,240.00. Total distribution cost climbs to Rs. 22,590.29.
2. **Fuel / Road Transportation Cost Surge (+25% and +50%):**
   - When transport rates increase by 25% (c_v → [23.1, 30.0, 39.4] Rs./km), transportation cost rises to Rs. 21,687.86. The optimizer responds by shifting border dealerships from f=1 to f=3 or f=5 to consolidate loads and reduce vehicle kilometers.
3. **Holding Cost Rate Inflation (+30%):**
   - As dealership storage space values increase in central prime zones (Nungambakkam, T. Nagar), the model forces tighter daily replenishments (f=1) to purge capital tied up in safety buffers.

9. Software Architecture, API Contracts & Dashboard UI

The decision-support system is deployed as an enterprise web architecture integrating a high-performance **FastAPI (Python 3.13)** backend with an interactive **Next.js 14** web application.

9.1 Backend REST API Endpoints

HTTP METHOD	ENDPOINT ROUTE	INPUT PAYLOAD	FUNCTIONAL DESCRIPTION & RESPONSE
POST	`/api/optimize`	`OptimizationInput`	Executes full preprocessing, calls solver (Gurobi or Heuristic), and returns routes, optimal frequencies, and cost breakdowns.
GET	`/api/forecast/sample`	None	Retrieves cached or executes on-demand XGBoost training, returning evaluation metrics, feature importances, and 7-day dealer forecasts.
POST	`/api/forecast`	None	Triggers XGBoost model retraining across historical demand records.
POST	`/api/forecast/upload`	`Multipart CSV File`	Accepts custom historical sales CSVs, validates schema, trains XGBoost, and generates customized forecasts.
GET	`/api/sample-data`	None	Provides baseline dealer, demand, and vehicle fleet configurations as JSON and base64 CSV.

9.2 Frontend Dashboard Architecture (Next.js 14)

The client interface is built with React 18, Next.js App Router, Tailwind CSS, Lucide Icons, and Leaflet OpenStreetMap. It features 6 dedicated analytics views:

- 📈 **XGBoost Demand Forecast View:** Displays model performance scorecards (RMSE, MAE, R^2, MAPE), interactive time-series charts with 7-day forward projections, feature importance rankings, and an "*Apply Forecast to Optimization*" action button.
- 📐 **Mathematical Model View:** Formally displays the project poster's reference formulations, sets, parameters, and all 8 MILP constraints.
- 🗺️ **Route Schedules & Leaflet Map:** Renders interactive polyline routes color-coded by vehicle across the Chennai street network.
- 🗓️ **Delivery Frequencies View:** Visualizes the optimal frequency y_{if}* assigned to each dealership and its scheduled visit calendar.
- 📦 **Safety Stock & Inventory View:** Provides stacked bar charts comparing cycle stock (d_i · f) and safety stock (SS_{if} = z · σ_i · √f), with a toggle to inspect the full candidate frequency matrix (f ∈ {1, 3, 5, 7}).
- 💰 **Executive Cost Summary:** Displays KPI financial scorecards, transport vs. holding cost ratios, and solver runtime metrics.

10. Strategic Recommendations, Managerial Insights & Roadmap

10.1 Key Managerial Insights for OEM Executives

1. **Eliminate Decentralized Ordering:** Dealerships often order inventory reactively based on local perceptions of shortage. Centralizing frequency and route planning from the Ambattur PDC unlocks immediate 15% to 25% cost reductions across the network.
2. **Exploit the Square-Root Law:** Consolidating low-volume dealerships into bi-weekly or weekly deliveries yields substantial transportation savings without causing proportional inventory blowouts, because safety stocks only increase with $\sqrt{f}$ rather than linearly.
3. **Leverage ML-Driven Dynamic Reorder Points:** Replacing static annual averages with dynamic XGBoost forecasts enables proactive pre-stocking before promotional rushes and prevents excess buffer accumulation during seasonal slowdowns.

10.2 Technical Limitations & Future Enhancement Roadmap

ENHANCEMENT AREA	CURRENT IMPLEMENTATION	FUTURE ENGINEERING ROADMAP (V3.0)
Geospatial Routing Engine	Haversine great-circle distance matrix	Integrate OSRM (Open Source Routing Machine) or Google Directions API for turn-by-turn road networks and live traffic congestion adjustments.
Stochastic VRP Recourse	Deterministic review-period normal demand	Formulate a Two-Stage Stochastic Program with Recourse or Chance-Constrained Programming to dynamically handle mid-cycle demand spikes.
Multi-Depot Topologies	Single Central PDC at Ambattur	Extend formulation to multi-depot networks with satellite stocking points in Tambaram and Sriperumbudur.
Fleet Electrification (EVs)	Combustion commercial vans (diesel)	Incorporate Electric Vehicle (EV) battery capacity, non-linear discharge rates, and charging window scheduling constraints.

Concluding Assessment:
The VRP-Inventory Optimizer provides an unprecedented bridge between cutting-edge operations research (MILP) and enterprise machine learning (XGBoost). By grounding logistics decisions in rigorous mathematical optimization, automotive OEMs can simultaneously achieve lean inventory operations and superior customer service delivery.